Abdallah Ahmed

Mechatronics Engineering • Class of 2028
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Submission Verification: This document provides direct access and technical dossier for my digital GitHub Engineering Portfolio, as required by the submission form (distinct from a standard CV).

Direct GitHub Portfolio Repository

Full source files, SolidWorks CAD, Altium Designer PCB projects, kinematic analysis, high-resolution visual documentation, and recorded lab demonstration videos are available on GitHub.

■ **Clickable Link:** https://github.com/3bdallahhahmed/Abdalla-Ahmed-portfolio

Scan with Phone

TECHNICAL PROJECTS SUMMARY (INCLUDED IN REPOSITORY)

1. Single-Actuator Gear-Driven Hexapod Walking Robot

E-JUST • MTR 221 Theory of Machines

Engineered a six-legged robot mimicking biological locomotion using single-motor mechanical synchronization via planar linkage synthesis (M=1 Grübler mobility) and an internal 90° bevel gear train.

■ `01-hexapod-walking-robot/ [CAD models, gear cutaway, foot trajectory, walking demo MP4]`

2. Industrial Robotics & Computer Vision Track

Electronics Research Institute (ERI)

60-hour intensive track at ERI Central Robotics Lab. Programmed ABB IRB 1200 & YuMi (14-DOF) cobots in RAPID; benchmarked DeepLabV3 vs FCN semantic segmentation; implemented YOLOv8 17-keypoint pose estimation.

■ `02-eri-robotics-simulation/ [ABB arms, segmentation comparisons, pose estimation, lab demo MP4]`

3. End-to-End PCB Design: Arduino Uno R3 & 5V Regulated Supply

Egyptian Space Agency (EgSA)

Comprehensive 60-hour hardware training in Altium Designer. Designed production-ready Arduino Uno R3 board (ATmega328P, USB UART, dual-rail mux) and 5V linear regulated supply with custom IPC 3D footprints.

■ `03-egsa-pcb-design/ [3D board renders, 2-layer routing layouts, complete circuit schematics]`

4. Autonomous Air Quality Monitoring Segway Robot

SoiLAB Eco-Warrior Camp • UAE

Two-wheeled self-balancing inverted pendulum robot with discrete PID feedback, 6-axis IMU sensor fusion (accelerometer + gyro), and real-time environmental particulate (PM2.5/PM10) and gas sampling payload.

■ `04-soilab-air-quality-robot/ [System dynamics, PID tuning documentation, hardware architecture]`

5. Etba3ly — 3D Printing & Prototyping Services

Maker Venture • etba3ly.store

Student-led rapid prototyping service providing precision FDM/resin manufacturing, engineering course prototypes, and scale display models (e.g. F-35 Lightning II scale model). Store: etba3ly.store

■ `05-etba3ly-3d-printing/ [F-35 scale model print documentation, service overview]`